Problem Set 2

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```r
## load data
df_data <- read_csv("ps2_data_spring26.csv", show_col_types = FALSE)
```

```r
glimpse(df_data)
```

```
## Rows: 2,500
## Columns: 8
## $ owner_older40       <dbl> 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0,~
## $ owner_male          <dbl> 0, 1, 1, 0, 1, 1, 1, 0, 0, 1, 0, 1, 0, 1, 1, 1, 1,~
## $ num_livestock       <dbl> 2, 2, 0, 5, 0, 7, 3, 1, 5, 1, 2, 2, 3, 3, 5, 2, 1,~
## $ years_edu           <dbl> 9, 10, 6, 8, 9, 4, 9, 9, 6, 8, 6, 4, 8, 10, 7, 5, ~
## $ num_employees       <dbl> 1, 3, 2, 1, 2, 3, 4, 1, 1, 2, 0, 3, 5, 1, 2, 1, 1,~
## $ grid_subsidy_amount <dbl> 150, 150, 200, 200, 50, 250, 100, 100, 50, 150, 20~
## $ grid_connection     <dbl> 0, 1, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ firm_revenue        <dbl> 11055.950, 19540.050, 10091.050, 6820.209, 10722.0~
```

```r
summary(df_data)
```

```
##  owner_older40      owner_male     num_livestock      years_edu     
##  Min.   :0.0000   Min.   :0.0000   Min.   : 0.000   Min.   : 4.000  
##  1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.: 1.000   1st Qu.: 5.000  
##  Median :0.0000   Median :0.0000   Median : 2.000   Median : 7.000  
##  Mean   :0.2484   Mean   :0.4996   Mean   : 2.354   Mean   : 6.967  
##  3rd Qu.:0.0000   3rd Qu.:1.0000   3rd Qu.: 3.000   3rd Qu.: 8.250  
##  Max.   :1.0000   Max.   :1.0000   Max.   :10.000   Max.   :10.000  
##  num_employees    grid_subsidy_amount grid_connection  firm_revenue  
##  Min.   : 0.000   Min.   : 50.0       Min.   :0.00    Min.   : 1720  
##  1st Qu.: 1.000   1st Qu.:100.0       1st Qu.:0.00    1st Qu.: 6764  
##  Median : 2.000   Median :150.0       Median :0.00    Median : 8889  
##  Mean   : 2.007   Mean   :150.1       Mean   :0.25    Mean   :11115  
##  3rd Qu.: 3.000   3rd Qu.:200.0       3rd Qu.:0.25    3rd Qu.:14477  
##  Max.   :10.000   Max.   :250.0       Max.   :1.00    Max.   :33958
```

1. The ideal experiment would be to randomly assign grid electricity connection to SMEs, and find the difference between the average firm annual revenues for SMEs who are assigned grid electricity and SMEs who are not assigned. We can calculate this difference by using the formula for the average treatment effect, which is as follows: tau ATE = E[Yi|Di=1] - E[Yi|Di=0] Di is whether SMEs are assigned to grid electricity connections, with 1 meaning the firm is connected and 0 meaning the firm is not connected. Yi is the outcome after being connected to the grid, which in this case would be a firm's annual revenue. E means that we take the average of SMEs' annual revenue across firms. We take two averages: the average firm annual revenue for firms who are connected to the electricity grid and the average firm annual revenue for firms who are not connected to the grid. The tau ATE is the difference in average revenues between SMEs who are connected to the grid and SMEs who are not connected.

2. If we simply compared connected to non-connected firms, then we wouldn't be accounting for fundamental differences between the firms, which would distort our calculation of the estimated effect of electricity connection. In other words, we would get the naive estimator, which does not account for selection bias.

The balance table below shows that the connected and non-connected firm groups are statistically different ($p>0.05$) on the number of livestock and the number of employees. This systemic difference between the two groups reduces my faith in the comparison that PEDAP wants to make because it shows that the two groups are meaningfully different on factors that could impact revenue.

```
# apply the regression to all pre-treatment covariates
regressions <- df_data %>%
  select("owner_older40",
         "owner_male",
         "num_livestock",
         "years_edu",
         "num_employees",
         "grid_subsidy_amount") %>%
  lapply(function(x) lm(x ~ grid_connection, data = df_data))

# pull the regression coefficient and p-value
balance_table <- regressions %>%
  sapply(function(x) coef(summary(x))[c(2,8)]) %>%
  t()

# format the table output
balance_table %>%
  kable(col.names = c("Differences in means", "p-value")) %>%
  kable_styling(position = "center", font_size = 8.5,
                latex_options = "hold_position")
```

	Differences in means	p-value
owner_older40	0.3856000	0.0000000
owner_male	0.1957333	0.0000000
num_livestock	0.0682667	0.4189095
years_edu	0.5882667	0.0000000
num_employees	0.1205333	0.0952956
grid_subsidy_amount	41.2800000	0.0000000

3. (optional, ungraded)

4. We decided to exclude the number of employees and number of livestock because their distribution was not significantly different across the treated and untreated groups. We saw the curse of dimensionality at play when certain grid-connected firms had no matching non-connected firms available for comparison. Our estimated average treatment effect (ATE) is 12,565, suggesting that the presence of grid connection contributes to a $12,565 annual revenue increase, on average, in Uganda. This estimated ATE is *less* than the naive estimator (12,992) by about $450/yr.

```
# check for common support for the conditional independence assumption
# make control variable
df_data <- df_data %>%
  mutate(no_grid_connection = ifelse(grid_connection==0,1,0))

# are there treated and control people at each level of the X variable?
df_data %>%
  group_by(num_employees) %>%
```

```r
  summarize(n_connected = sum(grid_connection),
            n_unconnected = sum(no_grid_connection),
            .groups = "keep") # to see total number of employee groups
```

```
## # A tibble: 11 x 3
## # Groups:   num_employees [11]
##    num_employees n_connected n_unconnected
##            <dbl>       <dbl>         <dbl>
##  1             0          83           308
##  2             1         175           545
##  3             2         151           423
##  4             3         114           279
##  5             4          53           199
##  6             5          29            69
##  7             6           9            32
##  8             7           8            16
##  9             8           2             3
## 10             9           1             0
## 11            10           0             1
```

```r
# check common support for multi-variable "cells" (check if we have treated and control observations ac
df_data %>%
  group_by(owner_older40, owner_male, num_livestock, num_employees) %>%
    summarize(n_connected = sum(grid_connection),
            n_unconnected = sum(no_grid_connection),
            .groups = "keep")
```

```
## # A tibble: 240 x 6
## # Groups:   owner_older40, owner_male, num_livestock, num_employees [240]
##    owner_older40 owner_male num_livestock num_employees n_connected
##            <dbl>      <dbl>         <dbl>         <dbl>       <dbl>
##  1             0          0             0             0           2
##  2             0          0             0             1           2
##  3             0          0             0             2           1
##  4             0          0             0             3           2
##  5             0          0             0             4           0
##  6             0          0             0             5           0
##  7             0          0             0             6           0
##  8             0          0             0             7           0
##  9             0          0             1             0           4
## 10             0          0             1             1           8
## # i 230 more rows
## # i 1 more variable: n_unconnected <dbl>
```

```r
# find the usable cells that we can find a treatment effect for
usable_cells <- df_data %>%
   group_by(owner_older40, owner_male, years_edu, grid_subsidy_amount) %>%
    summarize(n_connected = sum(grid_connection),
            n_unconnected = sum(no_grid_connection),
            .groups = "keep") %>%
  filter(n_connected > 0) %>%
  filter(n_unconnected > 0)

usable_cells
```

```
## # A tibble: 114 x 6
## # Groups:   owner_older40, owner_male, years_edu, grid_subsidy_amount [114]
##    owner_older40 owner_male years_edu grid_subsidy_amount n_connected
##            <dbl>      <dbl>     <dbl>               <dbl>       <dbl>
##  1             0          0         4                 100           1
##  2             0          0         4                 200           1
##  3             0          0         4                 250           1
##  4             0          0         5                 150           1
##  5             0          0         5                 200           4
##  6             0          0         5                 250           1
##  7             0          0         6                 150           2
##  8             0          0         6                 200           1
##  9             0          0         6                 250           5
## 10             0          0         7                 100           2
## # i 104 more rows
## # i 1 more variable: n_unconnected <dbl>
```

```r
# calculate the treatment effect

# calculate the means by cell * treatment status
grp <- df_data %>%
  group_by(owner_older40, owner_male, years_edu, grid_subsidy_amount) %>%
  summarize(mean_revenue_connected =
              sum(firm_revenue*grid_connection)/sum(grid_connection),
            mean_revenue_unconnected = sum(firm_revenue*no_grid_connection)/
              sum(no_grid_connection),
            count_connected = sum(grid_connection),
            count_unconnected = sum(no_grid_connection),
            .groups = "keep")

# convert the zero count cells to NA
grp <- grp %>% mutate(mean_revenue_connected = ifelse(count_connected==0, NA,
                                                      mean_revenue_connected),
                      mean_revenue_unconnected = ifelse(count_unconnected==0,NA,
                                                        mean_revenue_unconnected))

# take difference for each cell
grp$diff <- grp$mean_revenue_connected - grp$mean_revenue_unconnected
grp_fin  <- grp %>% filter(!is.na(diff))
```

```r
# estimate tau ATE as the weighted average
ate <- weighted.mean(grp_fin$diff,
                     w = (grp_fin$count_connected +
                            grp_fin$count_unconnected))
ate
```

```
## [1] 12564.58
```

```r
# calculate treatment effect using the naive estimator
naive <- mean(df_data$firm_revenue[df_data$grid_connection == 1]) -
  mean(df_data$firm_revenue[df_data$grid_connection == 0])

naive
```

```
## [1] 12992.22
```

```r
# estimate tau ATE as a weighted average
ate <- weighted.mean(grp_fin$diff,
                     w = (grp_fin$count_connected + grp_fin$count_unconnected))

ate
```

```
## [1] 12564.58
```

```r
# Estimate the tau_att as as weighted average
att <- weighted.mean(grp_fin$diff,
                     w = grp_fin$count_connected)
att
```

```
## [1] 11975.02
```

```r
# Find naive estimator

mean(subset(df_data, subset = grid_connection==1)$firm_revenue) -
  mean(subset(df_data, subset = grid_connection==0)$firm_revenue)
```

```
## [1] 12992.22
```

I decided to include all variables except for years of education and grid subsidy amount as these two variables had significant differences in means between the treatment and control groups. These characteristics were unbalanced between the treatment and control groups which could affect the calculated ATE. The calculation finds that SMEs who are connected to the grid have an increased 12111.93 dollars in firm annual revenue compared to SMEs who are not connected to the grid. The strength of this approach is now it is able to account for variables covariates to isolate the estimated treatment effect. The weakness of this approach is that the more covariates that are included in the calculation to be matched with each other, the less likely matches occur and the smaller the data set becomes. The result from the exact matching approach (12111.93) is lesser than the result from the naive estimator (12992.22).

5. Using instrumental variables (IV) would be a plausible strategy if the first stage (land gradients are related to being assigned a grid connection) and exclusion restriction (if land gradients only affect firm revenue through grid connections) requirements are met. The limitation of using an IV instrument is that it may be difficult to prove the exclusion restriction as it involves the error term of the equation. The exclusion restriction is satisfied by developing a convincing argument that the IV instrument only affects the outcome variable through the treatment. An alternative instrumental variable could be the type of cable that links SMEs to the national power grid. Power cables with higher voltage can affect the length of time required for charging or utilizing machinery, in turn affecting the firm's productivity and outputs which affects firm revenue. Power cables only affect firm revenue through connection to the grid. This would be a good instrument because power cable voltage remains constant and if provided by the government, does not face issues of spillovers or inaccurate reporting by SMEs.

6. We would use grid subsidy access as an instrumental variable for grid connection status We would first examine the impact of the subsidy on revenue. Then we would estimate the impact of the subsidy on a firm's likelihood of grid connection. We'd finally arrive at our estimated IV treatment effect by dividing the former estimate by the latter.

In math: Reduced form (regressing Yi on Zi): Yi = alpha + theta*Zi + Xi + ni where i is an individual firm, Zi equals 1 if i received the subsidy and 0 otherwise, Xi represents covariates, and ni represents the error term.

First stage (regressing Di on Zi): Di = alpha + gamma*Zi + ßXi + ni where Di is 1 if i is connected to the grid and 0 otherwise, and Zi, Xi, and ni have the same meaning as in the previous formula.

IV estimate = theta/gamma

7.

```r
# run first stage regression of grid subsidy amount on grid connection
first_stage <- lm(grid_connection ~ grid_subsidy_amount + years_edu,
                  data=df_data)

summary(first_stage)
```

```
##
## Call:
## lm(formula = grid_connection ~ grid_subsidy_amount + years_edu,
##     data = df_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.5091 -0.2855 -0.1655  0.1280  0.9725
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)         -0.2223047  0.0377517  -5.889 4.42e-09 ***
## grid_subsidy_amount  0.0015452  0.0001173  13.175  < 2e-16 ***
## years_edu            0.0345062  0.0046517   7.418 1.62e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4147 on 2497 degrees of freedom
## Multiple R-squared:  0.08409,    Adjusted R-squared:  0.08335
## F-statistic: 114.6 on 2 and 2497 DF,  p-value: < 2.2e-16
```

This estimates the variation in grid connection take-up caused by the grid subsidy amount. The estimate finds that 0.0015452 percentage points of the variation in grid connection take-up is caused by the grid subsidy amount.

8. This regression estimates the impact of grid subsidy access (not actual grid connection) on revenue, a.k.a. the reduced form estimator. This parameter has limited use for policy. On the one hand, policymakers should care about how the subsidy impacts revenue. On the other hand, selection bias could distort the estimate because the subsidy might only be used by firms who would have obtained grid connection anyway. Essentially, this is a useful parameter in concept, but the aforementioned regression is an unreliable way to estimate it. The regression produces a coefficient of 12.90 ($p<0.05$), suggesting that the subsidy increases annual revenue by about $12.90 per year on average.

```r
# run regression (reduced form)

reg_subsidy <- lm(df_data$firm_revenue ~ df_data$grid_subsidy_amount + df_data$owner_older40 + df_data$
                    df_data$years_edu + df_data$num_employees)

summary(reg_subsidy)
```

```
##
## Call:
## lm(formula = df_data$firm_revenue ~ df_data$grid_subsidy_amount +
##     df_data$owner_older40 + df_data$owner_male + df_data$num_livestock +
##     df_data$years_edu + df_data$num_employees)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -9870  -3415  -1193   2369  16831
```

```
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)                  816.50     488.57   1.671  0.09481 .
## df_data$grid_subsidy_amount   12.90       1.37   9.420  < 2e-16 ***
## df_data$owner_older40       6910.62     224.22  30.821  < 2e-16 ***
## df_data$owner_male          3271.32     193.57  16.900  < 2e-16 ***
## df_data$num_livestock         52.55      52.95   0.992  0.32109
## df_data$years_edu            648.02      54.34  11.925  < 2e-16 ***
## df_data$num_employees        185.84      61.93   3.001  0.00272 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4836 on 2493 degrees of freedom
## Multiple R-squared:  0.3688, Adjusted R-squared:  0.3673
## F-statistic: 242.8 on 6 and 2493 DF,  p-value: < 2.2e-16
```

9.

```r
# run first stage regression of grid subsidy amount on grid connection
first_stage <- lm(grid_connection ~ grid_subsidy_amount + years_edu,
                  data=df_data)

# run second stage regression of grid connection on firm revenue
second_stage <- lm(firm_revenue ~ first_stage$fitted.values + years_edu,
                   data = df_data)

summary(second_stage)
```

```
##
## Call:
## lm(formula = firm_revenue ~ first_stage$fitted.values + years_edu,
##     data = df_data)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -9964  -4305  -2003   3937  24162
##
## Coefficients:
##                           Estimate Std. Error t value Pr(>|t|)
## (Intercept)                6376.14     476.73  13.375  < 2e-16 ***
## first_stage$fitted.values  7409.70    1081.50   6.851 9.18e-12 ***
## years_edu                   414.30      76.17   5.439 5.87e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5908 on 2497 degrees of freedom
## Multiple R-squared:  0.05651,    Adjusted R-squared:  0.05576
## F-statistic: 74.78 on 2 and 2497 DF,  p-value: < 2.2e-16
```

```r
# run iv reg using ivreg
iv_reg <- AER::ivreg(firm_revenue ~ grid_connection + years_edu |
                       grid_subsidy_amount + years_edu, data = df_data)
summary(iv_reg)
```

```
##
```

```
## Call:
## AER::ivreg(formula = firm_revenue ~ grid_connection + years_edu |
##     grid_subsidy_amount + years_edu, data = df_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -6530.7 -2515.5  -284.7  2004.4 18100.8
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)       6376.1      262.9  24.253   <2e-16 ***
## grid_connection   7409.7      596.4  12.424   <2e-16 ***
## years_edu          414.3       42.0   9.863   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3258 on 2497 degrees of freedom
## Multiple R-Squared: 0.7131,  Adjusted R-squared: 0.7128
## Wald test: 245.9 on 2 and 2497 DF,  p-value: < 2.2e-16
```

```r
# alternate method (non-2SLS):
# I also messed around with the covariates here FYI

# Step 1: calculate the reduced form (as in Question 8)

reg_subsidy <- lm(df_data$firm_revenue ~ df_data$grid_subsidy_amount + df_data$owner_older40 + df_data$
                    df_data$years_edu + df_data$num_employees
                   )

summary(reg_subsidy) # reduced form est = 12.90
```

```
##
## Call:
## lm(formula = df_data$firm_revenue ~ df_data$grid_subsidy_amount +
##     df_data$owner_older40 + df_data$owner_male + df_data$num_livestock +
##     df_data$years_edu + df_data$num_employees)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -9870  -3415  -1193   2369  16831
##
## Coefficients:
##                             Estimate Std. Error t value Pr(>|t|)
## (Intercept)                   816.50     488.57   1.671  0.09481 .
## df_data$grid_subsidy_amount    12.90       1.37   9.420  < 2e-16 ***
## df_data$owner_older40        6910.62     224.22  30.821  < 2e-16 ***
## df_data$owner_male           3271.32     193.57  16.900  < 2e-16 ***
## df_data$num_livestock          52.55      52.95   0.992  0.32109
## df_data$years_edu             648.02      54.34  11.925  < 2e-16 ***
## df_data$num_employees         185.84      61.93   3.001  0.00272 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4836 on 2493 degrees of freedom
## Multiple R-squared:  0.3688, Adjusted R-squared:  0.3673
```

```
## F-statistic: 242.8 on 6 and 2493 DF,  p-value: < 2.2e-16
```

```r
# Step 2: calculate the first stage estimate (as in Question 9)
first_stage <- lm(grid_connection ~ grid_subsidy_amount + years_edu + owner_older40 + owner_male + num_l
                  data=df_data)

summary(first_stage) # first stage estimate = 0.0016333
```

```
##
## Call:
## lm(formula = grid_connection ~ grid_subsidy_amount + years_edu +
##     owner_older40 + owner_male + num_livestock + years_edu +
##     num_employees, data = df_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.74734 -0.25223 -0.09156  0.16344  1.09440
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)         -0.4207487  0.0374002 -11.250  < 2e-16 ***
## grid_subsidy_amount  0.0016333  0.0001048  15.579  < 2e-16 ***
## years_edu            0.0330955  0.0041597   7.956 2.67e-15 ***
## owner_older40        0.3978959  0.0171641  23.182  < 2e-16 ***
## owner_male           0.1507493  0.0148181  10.173  < 2e-16 ***
## num_livestock        0.0011636  0.0040532   0.287   0.7741
## num_employees        0.0090467  0.0047405   1.908   0.0565 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3702 on 2493 degrees of freedom
## Multiple R-squared:  0.271,  Adjusted R-squared:  0.2692
## F-statistic: 154.5 on 6 and 2493 DF,  p-value: < 2.2e-16
```

```r
# Step 3: Divide reduced form / first stage

12.90 / 0.0016333 # final estimate: 7898.12
```

```
## [1] 7898.12
```

Grid electricity connection does matter for firm revenues. Being connected to the electricity grid increases firm revenues by $7409.7. This finding is statistically significant, with a p-value of <2e-16. The standard error is 596.4, which does not cause concern because it is small relative to the coefficient estimate of 7409.7 and it's smaller than the residual standard deviation of 3258.

10. Random noise associated with the subsidy amount data would constitute a classical measurement error in Di. This causes attenuation bias (it biases the causal estimate towards 0) because of the bias that is embedded in the treatment (see Lecture 8, slides 16-17). Luckily, the use of instrumental variables can correct the bias.

A systemic error in the subsidy amount data (non-classical measurement error in Di) would also bias our estimate, and instrumental variables could not address the problem.